

DATABASE OPTIMIZATION STRATEGY

Oracle Query Rewrite Monitoring

Preventing Silent Performance Degradation in High-Integrity Environments

The Silent Performance Trap

Why Staleness is the Leading Cause of Unexpected I/O Spikes

| INTEGRITY VS. PERFORMANCE

ENFORCED Integrity

The default setting (`query_rewrite_integrity = ENFORCED`). Oracle guarantees 100% data freshness. Any stale Materialized View is immediately discarded by the Optimizer.

The Silent Result

Queries silently fall back to scanning base tables. No ORA-errors are raised, leading to massive resource consumption that stays hidden until production performance dips.

| IMPACT OF DATA STALENESS

When base table DML occurs, the MV status changes to **STALE**. Under high-integrity settings, the rewrite engine essentially "blinds" itself to the MV.

- ✓ Millions of rows are processed instead of thousands.
- ✓ Aggregate results are re-calculated on-the-fly.
- ✓ Memory and Buffer Cache pressure increases.



| KEY MONITORING METRICS



STALENESS

Monitors sync status between MV and its base tables. Values: FRESH, STALE, UNUSABLE.



COMPILE_STATE

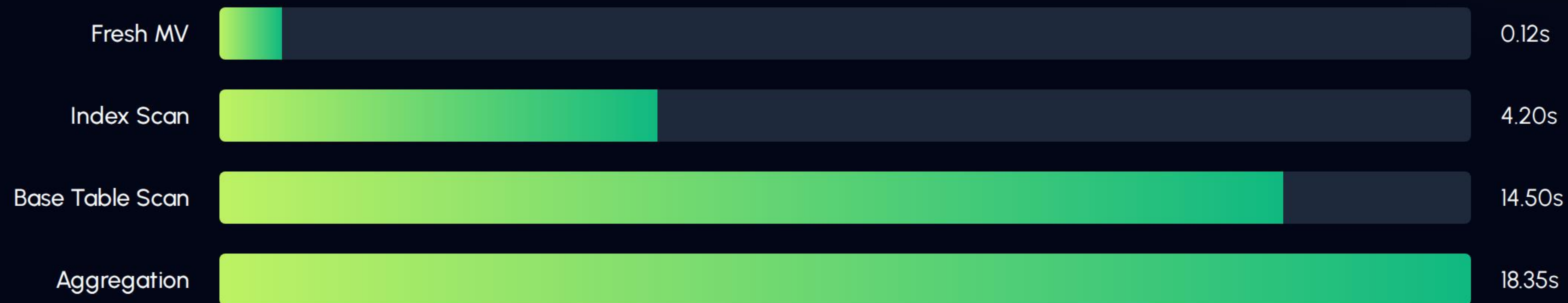
Validates if the MV metadata matches the current table definitions.



REWRITE_ENABLED

Flags if the MV is eligible for optimizer utilization.

PERFORMANCE COMPARISON



*Silent bypass of a stale MV can increase query latency by over **15,000%** in large datasets.*

Implementation & SQL Demo

Building a Robust Monitoring Framework in USER_MVIEWS

CRITICAL MONITORING QUERIES

Target Metric	SQL Query Snippet
Stale MVs Filter	<code>SELECT mview_name, last_refresh_date FROM user_mviews WHERE staleness = 'STALE';</code>
Performance Risk	<code>SELECT owner, mview_name FROM all_mviews WHERE query_rewrite_enabled = 'Y' AND staleness != 'FRESH';</code>
State Check	<code>SELECT mview_name, compile_state FROM user_mviews WHERE compile_state != 'VALID';</code>

| ALERTING BEST PRACTICES

- ✓ **Intelligent Filtering:** Only trigger high-priority alerts for MVs where `QUERY_REWRITE_ENABLED = 'Y'`.
 - ✓ **Sampling Frequency:** Poll the `USER_MVIEWS` dictionary every 15-30 minutes to detect drift.
 - ✓ **Grace Periods:** Allow a buffer for planned batch refreshes before escalating "Stale" alerts to P1 severity.
 - ✓ **Visual Dashboards:** Integrate staleness metrics into operational dashboards (Grafana/OEM) to correlate with CPU spikes.
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CPU LOAD CORRELATION



Silent Fallback: System CPU Load jumps from 15% to 80% as rewrite fails.

REMEDIATION WORKFLOW



Maintain Peak Performance

Don't let staleness degrade your database health silently. Implementing robust dictionary monitoring ensures your optimized query paths remain intact.

www.oracle-optimization.com

| IMAGE SOURCES



<https://media.istockphoto.com/id/912117190/vector/notification-laptop-and-speech-bubble-with-exclamation-mark-icon-on-screen-computer-message.jpg?s=170667a&w=is&k=20&c=cFA9XubbmgeeeCSBE6VM76yvA3brJS7dwzQMv9T22dg=>

Source: www.istockphoto.com



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Source: www.vecteezy.com
